Workshop 4 - Dashboard Implementation Report for NYC Gastronomic Trends and Sentiment Analysis

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***Abstract*—This report presents the implementation of two dashboard applications for the Empire's Taste project, a Medallion-style data pipeline for analyzing NYC food media trends, recipe availability, and public sentiment. The workshop focused on end-to-end integration from Airflow orchestration to Gold-layer Parquet outputs and on the construction of independent governance and storytelling dashboards with Plotly Dash. The final result is a reproducible visualization layer in which the governance dashboard monitors data quality and pipeline health, while the storytelling dashboard translates sentiment, keyword, entity, source, and activity trends into insights for food professionals.**

***Index Terms*—Apache Airflow, data dashboards, data governance, Gold layer, Medallion architecture, Plotly Dash, sentiment analysis, visualization design**

I. INTRODUCTION

Food professionals and gastronomic media analysts increasingly need tools that connect public discussion, recipe availability, and data quality into a single operational view. The Empire's Taste project addresses this need by processing NYC food-media content and recipe information through a Medallion-style data pipeline. Apache Airflow is used to orchestrate the end-to-end workflow, while PySpark supports the transformation logic required to prepare analytical Gold outputs for consumption by downstream applications [1], [2].

The main problem addressed in this workshop was not only to create visualizations but to ensure that they were connected to a reliable and reproducible data pipeline. A dashboard that reads incomplete, outdated, or schema-incompatible files can mislead users even if its charts are visually correct. For that reason, this report focuses on the operational link between Bronze ingestion, Silver cleaning, Gold aggregation, and the two browser-based dashboards. The presentation layer reads validated Parquet outputs with Pandas, reducing direct dependence on raw data, external APIs, or intermediate Silver schemas [3].

Two dashboard perspectives were implemented. The governance dashboard serves the data engineering team by exposing quality indicators such as record volume, null rates, duplicate rates, outlier rates, and schema compliance. The storytelling dashboard serves the functional user by presenting sentiment distribution, sentiment trends, top food keywords, named entities, source comparison, and activity volume in plain language. The sentiment and entity components align with the project use of VADER-style sentiment scoring and spaCy named-entity extraction [4], [5]. Both interfaces were implemented with Plotly Dash and Plotly visual components, which support interactive browser-based analytical applications [6], [7].

This report documents the integration adjustments, dashboard layouts, KPI and user-story mappings, visualization design decisions, reproducibility evidence, and challenges encountered during implementation. It also distinguishes the work completed in this workshop from earlier data preparation improvements by focusing on dashboard readiness, Gold-layer compatibility, browser accessibility, and repeatable execution in the Docker-based project environment [8].

II. INTEGRATION ADJUSTMENTS

The integration work in this workshop focused on validating that the dashboard-ready Gold layer could be regenerated from a clean state and then consumed independently by the two presentation applications. The complete Medallion flow was executed inside Docker through Apache Airflow, starting with Bronze ingestion, continuing with Silver cleaning and keyword extraction, and finishing with the Gold PySpark transformation. In the final implementation, the handoff between layers is enforced through explicit Airflow task dependencies in a single deterministic sequence:

`fetch_eater_ny` → `clean_eater_ny` → `extract_trending_keywords` → `fetch_recipes` → `clean_recipes` → `prepare_gold`.

Table I summarizes the verified pipeline flow in a compact way, grouping the six Airflow tasks by their integration role.

This verification confirmed that each downstream step only runs after the required upstream files are available, replacing fragile manual execution with an end-to-end reproducible pipeline. The check also validated that all outputs are written to date-partitioned folders, allowing the dashboards to select the latest available Gold partition without depending on Bronze or Silver files.

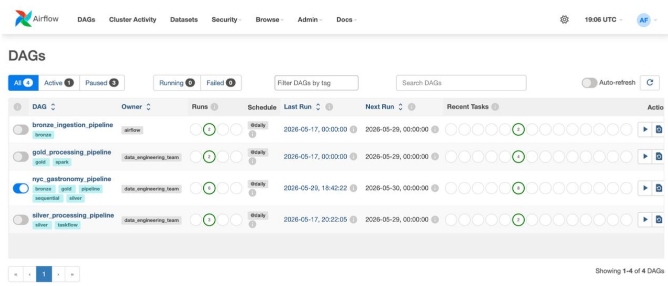

Fig. 1. Airflow DAG showing the end-to-end Medallion pipeline task sequence.

TABLE I
CONCISE END-TO-END PIPELINE VERIFICATION.

Stage	Layer	Verified Output
Article ingestion	Bronze	Raw Eater NY article files
Article cleaning and keywords	Silver	Clean article Parquet and food keywords
Recipe ingestion and cleaning	Bronze/Silver	Spoonacular raw and flattened recipe data
Dashboard preparation	Gold	Article, recipe, governance, and storytelling Parquet files

Several integration corrections were required before the dashboards could read the Gold layer reliably. First, the Docker startup procedure was adjusted to use `docker-compose up --build` when rebuilding the Airflow environment, because recreating containers without rebuilding reused an image that did not include PySpark. Second, environment updates such as the Spoonacular API key were applied with `docker-compose up -d --force-recreate`, ensuring that containers reload the current `.env` configuration instead of keeping an outdated runtime state. Third, the Gold aggregation logic was corrected in the keyword sentiment summary: Spark row objects were accessed through attributes instead of dictionary-style indexing, resolving the `_FakeRow` access error that interrupted the `agg_keyword_sentiment()` computation.

Table II condenses the integration issues into short correction groups. The table is intentionally concise; the surrounding paragraphs explain the operational impact of each adjustment.

The Gold files were then checked for dashboard completeness. The governance output, named with the pattern `governance_*.parquet`, contains the KPI rows required by the engineering dashboard, including completeness, volume, uniqueness, validity, duplicate-rate, and schema-compliance indicators. The storytelling output, named with the pattern `storytelling_*.parquet`, contains the aggregations required by the functional dashboard, including sentiment distribution, keyword sentiment, weekly trend, source comparison, and named-entity summaries. These checks ensured that the Gold layer exposes the fields expected by both dashboards and that no presentation component depends on intermediate Silver schemas.

The main distinction from the previous workshop is that

TABLE II
CONCISE INTEGRATION CORRECTIONS.

Issue Area	Workshop 4 Adjustment
Container image	Rebuilt services so PySpark and NLP dependencies were available
Runtime configuration	Recreated containers to reload the current API key and `.env` values
Gold aggregation	Corrected Spark Row access in the keyword sentiment summary
Keyword quality	Added food-domain filtering to remove generic non-culinary terms

Workshop #3 addressed data preparation and analytical improvements, while this workshop concentrated on operational integration and dashboard readiness. Therefore, the adjustments reported here are limited to pipeline execution, container rebuilding, configuration reloading, Gold aggregation compatibility, and verification that both dashboards can start independently from their corresponding Gold Parquet files without runtime errors.

III. GOVERNANCE DASHBOARD DOCUMENTATION

The governance dashboard was designed as an operational monitoring interface for the data engineering team. Its purpose is to make the current quality state of the Medallion pipeline visible without requiring users to inspect Parquet files manually. The application is implemented in `dashboard/governance_app.py`, runs at `localhost:8050`, and loads the latest `governance_*.parquet` file with `pd.read_parquet()`. This design keeps the dashboard decoupled from the Bronze and Silver layers and guarantees that the interface reflects only the validated Gold governance output.

A. *Layout Description and Components*

The layout follows a top-down monitoring workflow. The header identifies the project, the dashboard purpose, and the latest data refresh time. Immediately below the header, four KPI summary cards provide a quick health overview of the pipeline: total records, maximum null rate, duplicate rate, and schema compliance rate. These cards are placed first because they answer the most important governance question: whether enough data arrived and whether it is usable for analysis.

The second visual layer contains the detailed diagnostic charts. A horizontal null-rate bar chart ranks fields by missing-value severity so that incomplete attributes can be identified quickly. A volume-over-time chart compares ingested records by date and source, helping detect drops in either the web scraping or API ingestion branches. An outlier-rate bar chart summarizes the proportion of outliers in numeric fields and highlights values that exceed the accepted threshold. Finally, the bottom section contains a data quality summary table with the complete list of governance KPIs, including category, source, field, value, unit, and finding.

TABLE III
CONCISE GOVERNANCE KPI GROUPS.

Dashboard Area	Gold KPI Inputs
Summary cards	Core volume, completeness, duplication, and schema metrics
Quality charts	Null and outlier rates by field
Volume chart	Record count by date and source
Audit table	Complete governance KPI dataset

B. KPI Mapping

Each dashboard component is mapped directly to governance metrics produced in the Gold layer. Table III keeps the mapping concise by grouping related dashboard elements instead of listing every card and chart as a separate row.

The summary cards condense the most important Gold governance indicators into a single row of immediate status signals. Total records are calculated from `record_count` rows to show whether the ingestion process produced the expected volume. The maximum null-rate card scans the field-level `null_rate` values and surfaces the worst completeness issue, while the duplicate-rate card reports whether repeated records are affecting the data. The schema-compliance card summarizes whether the current Gold output still follows the structure expected by the dashboards.

The diagnostic charts expand those high-level signals into actionable detail. The null-rate chart uses field-level completeness KPIs and orders them by severity so engineers can identify which attributes require attention first. The outlier-rate chart focuses on numeric fields and compares their anomaly rate against the accepted threshold. The volume chart uses `record_count` aggregated by date and source to reveal whether the API branch or web-scraping branch produced an unexpected drop. Finally, the audit table preserves all Gold governance rows so the team can inspect the exact category, source, field, value, unit, and finding behind each visual summary.

C. Design Rationale

The governance dashboard uses a professional monitoring layout based on progressive detail: summary cards first, diagnostic charts second, and a full audit table last. This arrangement reduces cognitive load because users can first determine whether the pipeline is healthy and then investigate the source of a problem only when needed. The horizontal orientation of the null-rate chart was selected because field names can be long, and ordering bars by severity makes missing-data issues easier to compare. The volume chart uses grouped bars or time-based traces to compare sources across dates, which is appropriate for detecting ingestion failures or sudden drops in coverage. The outlier chart uses vertical bars with a threshold reference because outlier rates are field-level numeric diagnostics that benefit from direct comparison.

The color scheme is shared with the rest of the project to preserve visual consistency. Dark blue is used for primary structural elements, amber for secondary emphasis, and severity colors are reserved for data quality status: yellow for low null rates, orange for medium null rates, and red for high-risk values. The dashboard also includes an automatic refresh mechanism using `dcc.Interval` every five minutes, so the displayed metrics update when a new Gold partition is produced. If no governance Gold file is available, the application shows a warning instead of failing, which improves reliability during development and deployment.

IV. STORYTELLING DASHBOARD DOCUMENTATION

The storytelling dashboard was designed for the functional user profile defined for the project: NYC chefs, food professionals, and gastronomic media analysts. Unlike the governance dashboard, this interface does not emphasize pipeline health or technical metadata. Its goal is to translate the Gold analytical outputs into a readable story about what food topics are being discussed, how people feel about them, and how those discussions relate to recipe availability. The application is implemented in `dashboard/storytelling_app.py`, runs at `localhost:8051`, and loads the latest `storytelling_*.parquet` file through `pd.read_parquet()`. The resulting Pandas DataFrame feeds the Dash components used in the visual interface.

A. Layout Description and Components

The layout follows a narrative path from overview to detail. It starts with a header that displays the project title and the latest update timestamp. The first content element is a narrative summary card written in plain language. It combines the main food topic, the share of positive coverage, recipe availability, and the most relevant dietary signal into a short conclusion that can be understood without reading any chart.

After the summary, the dashboard presents a contextual mentions card that lists the most referenced restaurants, neighborhoods, and people extracted from the articles. This card helps users recognize who and where the conversation is about before moving into quantitative charts. The next row contains the sentiment distribution donut chart and the sentiment trend line chart. The donut chart shows the overall split between positive, neutral, and negative records, while the trend chart shows whether the average sentiment is improving, declining, or remaining stable over time.

The lower section provides topic and source-level detail. A horizontal top keywords bar chart ranks the most relevant food terms and colors each bar by its associated sentiment. A source comparison chart places scraped media coverage and API recipe availability side by side, helping users identify whether recipes exist for the topics being discussed in NYC food media. A volume activity chart closes the flow by showing record counts per time period, making peaks of public discussion easier to identify.

The implemented component list includes:

- **Narrative Summary Card:** plain-language conclusion for the current data window.
- **Sentiment Distribution Donut:** positive, neutral, and negative record breakdown.

TABLE IV
CONCISE STORYTELLING USER-STORY MAP.

Story Stage	Dashboard Support
Immediate takeaway	Narrative summary card
Mood over time	Sentiment donut and trend line
Topics and context	Top keywords and NYC mentions cards
Supply and activity	Source comparison and volume activity charts

- **Sentiment Trend Line:** average sentiment movement by time period.
- **Top Keywords Bar Chart:** ranked food terms colored by associated sentiment.
- **Source Comparison Chart:** side-by-side comparison between API and scraping sources.
- **Volume Activity Chart:** record volume by period to show discussion peaks.
- **NYC Mentions Card:** restaurants, neighborhoods, and people extracted from articles.

B. User Story Mapping

The dashboard components were selected according to the decisions a food professional needs to make when interpreting public opinion. Table IV presents a compact user-story map, while the paragraphs after the table explain how each stage supports the narrative.

The immediate takeaway stage maps to the user need of understanding the main finding before interacting with the charts. The narrative summary card provides this answer in one plain-language paragraph by combining the leading topic, the percentage of positive coverage, available recipes, and the most relevant diet signal. This allows a chef or media analyst to understand the week's conclusion without interpreting technical scores.

The mood-over-time stage answers two related questions: what is the current tone of the conversation, and whether that tone is changing. The sentiment donut chart shows the current positive, neutral, and negative distribution at a glance. The trend line adds temporal context by showing average sentiment by period, so the user can distinguish a stable public mood from a recent shift in opinion.

The topics-and-context stage explains what the conversation is about. The top keywords chart identifies the food terms that dominate the corpus and colors them by associated sentiment, making it possible to separate popular positive topics from topics with negative coverage. The NYC mentions card adds a more recognizable layer by listing restaurants, neighborhoods, and people mentioned in the articles, which helps users connect abstract keyword trends to real places and actors in the local gastronomy scene.

The supply-and-activity stage connects media attention with practical action. The source comparison chart contrasts article volume with recipe availability, helping users see whether the recipe database supports the topics currently being discussed. The volume activity chart shows when public discussion is most active, making peaks in coverage easier to identify and interpret.

C. Narrative Rationale

The dashboard is organized as a guided story instead of a collection of independent charts. The first screen gives the conclusion, the middle explains sentiment and topic direction, and the final charts provide supporting detail about sources and activity volume. This order prioritizes insight over raw numbers, which is appropriate for chefs and gastronomic media analysts who need fast interpretation rather than technical diagnostics.

Plain-language labels are used throughout the interface. Technical expressions such as compound sentiment score or TF-IDF are avoided in visible labels, even though those metrics are used internally to build the Gold layer. The donut chart was selected for the sentiment distribution because it provides an immediate high-level split and can emphasize the positive-coverage percentage in the center. The sentiment trend uses a line chart because opinion changes are sequential, while the keyword chart uses horizontal bars because food terms can be long and are easier to read on a vertical list.

The interface also supports interactive exploration through Plotly hover, zooming, and chart filtering. Hover text gives additional context without crowding the main view, zooming allows users to inspect specific time windows, and source-level comparisons help connect public discussion with recipe availability. The dashboard refreshes from the latest Gold partition so that the story remains aligned with the most recent successful pipeline run.

V. VISUALIZATION DESIGN DECISIONS

The visualization layer was designed to keep both dashboards visually coherent while serving different users. The governance dashboard prioritizes operational monitoring and fast anomaly detection, whereas the storytelling dashboard prioritizes interpretation for chefs, food professionals, and gastronomic media analysts. Even with these different goals, both dashboards follow the same quality standards: every chart includes a descriptive title, labeled axes with units where applicable, legends when multiple series are shown, and a source annotation indicating that the data comes from the latest validated Gold Parquet files.

A. Chart Type Selection

Chart types were selected according to the analytical task rather than visual novelty. Table V summarizes the main decisions in a compact form, while the text below explains the rationale.

Numeric cards were used for headline governance indicators because totals, maximum rates, and compliance values must be read immediately without comparing many marks. Horizontal bar charts were selected for null rates and top keywords because both views contain text-heavy labels; placing labels on the vertical axis keeps them readable and allows the bars to be ordered by severity or relevance. Grouped bar charts were used when the goal was comparison between sources, such as API data versus web-scraped articles, because side-by-side bars make differences in volume easier to identify.

TABLE V
CONCISE CHART TYPE DECISIONS.

Analytical Need	Selected Visualization
Single KPI status	Numeric summary cards
Field or keyword ranking	Horizontal bar charts
Source comparison	Grouped bar charts
Time evolution	Line charts with markers
Three-class sentiment split	Donut chart
Detailed inspection	Sortable data table

TABLE VI
CONCISE SHARED COLOR PALETTE.

Visual Role	Encoding
Positive sentiment	Green `#24a058`
Negative sentiment	Red `#da3636`
Neutral sentiment	Yellow `#f5c518`
Primary structure	Black `#0d0d0f` backgrounds and headers
Text	Cream `#eeeae2` labels and body text

Line charts were reserved for time series, especially sentiment trends, because they emphasize continuity and direction across daily or weekly periods. The only pie-like visualization is the sentiment donut chart, and it is used only because sentiment has three categories: positive, neutral, and negative. For comparisons with more than three categories, the dashboards avoid pie charts and use bars instead, preventing the visual distortion that appears when users must compare many slices. The data quality table remains necessary because it gives engineers a complete audit view after the charts identify where to investigate.

B. Color Palette

A shared palette is used across both applications so that the same visual language means the same thing in every dashboard. Table VI shows the palette roles without overloading the report with implementation details.

The sentiment encoding is consistent in every sentiment-related chart: green (`#24a058`) represents positive records, red (`#da3636`) represents negative records, and yellow (`#f5c518`) represents neutral values. This convention is applied to the sentiment donut, keyword sentiment bars, and sentiment trend annotations. For governance quality, severity colors follow the same red encoding for high-risk values, reserving yellow-neon for medium-risk warnings and cream tones for low-risk indicators. To support accessibility, color is not the only source of meaning. Labels, legends, values, hover text, and chart titles also communicate the interpretation, which makes the dashboards easier to read for colorblind users.

C. Layout and Legibility

Both dashboards use a professional top-down layout. Each page begins with a header containing the project title, topic angle, team name, and the latest data update timestamp. This header makes the dashboard self-contained for screenshots and reproducibility evidence because it identifies what the user is viewing and when the Gold data was last refreshed.

The body layout is organized from summary to detail. KPI cards and narrative cards appear first because they communicate the main conclusion quickly. Charts are then arranged in responsive rows so that a standard 1080p screen can display the dashboard without horizontal scrolling. Wider diagnostic elements, such as summary tables or comparison charts, are placed below the main overview so they can use more horizontal space without crowding the first screen.

Each chart follows the same annotation standard. Titles state the question being answered, axes describe the measured variable and include units such as percent or record count when applicable, legends are shown when multiple categories or sources appear in the same plot, and source annotations identify the Gold data used by the visualization. These decisions make the dashboards readable as standalone analytical artifacts rather than only as interactive web pages.

VI. REPRODUCIBILITY EVIDENCE

Reproducibility was validated by starting the complete Docker environment from a clean state, rebuilding the services when dependencies changed, triggering the Airflow pipeline, and then opening both dashboards in a browser. The expected startup sequence is: create or update the `.env` file with the Spoonacular API key, run `docker-compose up --build -d`, access the Airflow web interface at `localhost:8080`, trigger the `nyc_gastronomy_pipeline` DAG, and wait until the Gold preparation step finishes successfully. After that point, the governance dashboard must be available at `localhost:8050` and the storytelling dashboard at `localhost:8051`.

The evidence required for this workshop consists of three browser screenshots: one showing the full pipeline execution in the Airflow UI and one for each running dashboard. The Airflow screenshot should show the DAG run completed or running through the expected task sequence, including Bronze ingestion, Silver processing, recipe ingestion, recipe cleaning, and Gold preparation. The dashboard screenshots should show that each application opens independently and loads its corresponding Gold Parquet data without errors.

A. Required Workshop Evidence

The three figures in this subsection correspond to the evidence explicitly requested by Workshop 4: the full Airflow pipeline execution, the governance Dash dashboard, and the storytelling Dash dashboard. These screenshots should be inserted as final evidence once the images are available.

TABLE VII
CONCISE REPRODUCIBILITY EVIDENCE CHECKLIST.

Evidence	What It Demonstrates
Airflow UI	Complete pipeline execution and task dependency order
Governance dashboard	Access to `governance_*.parquet` through the browser
Storytelling dashboard	Access to `storytelling_*.parquet` through the browser
Gold-only loading	Dashboards read validated Gold outputs, not Bronze or Silver files

Fig. 2. Evidence of the full Medallion pipeline running in Apache Airflow.

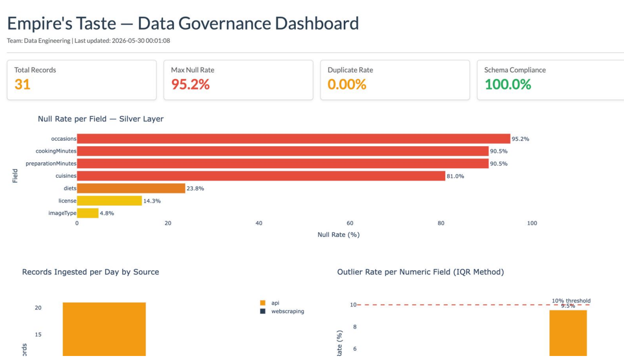

Fig. 3. Evidence of the governance dashboard reading the Gold governance Parquet output.

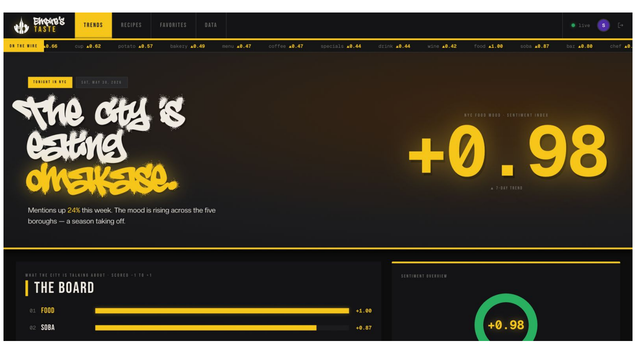

Fig. 4. Evidence of the storytelling dashboard reading the Gold storytelling Parquet output.

B. Supplementary Web and API Evidence

The following placeholders are supplementary because they document the broader web application and API integration around the dashboards. They support the project narrative, but they do not replace the three required Workshop 4 screenshots listed above.

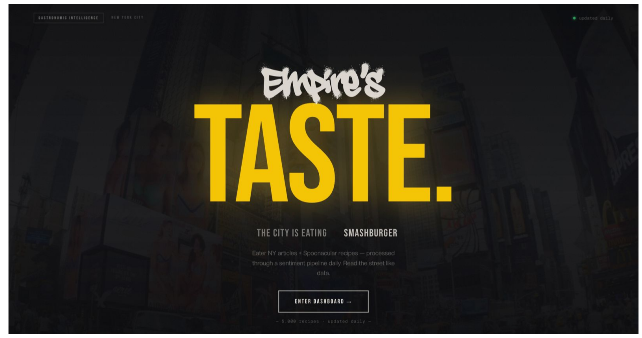

Fig. 5. Supplementary evidence of the web application landing page.

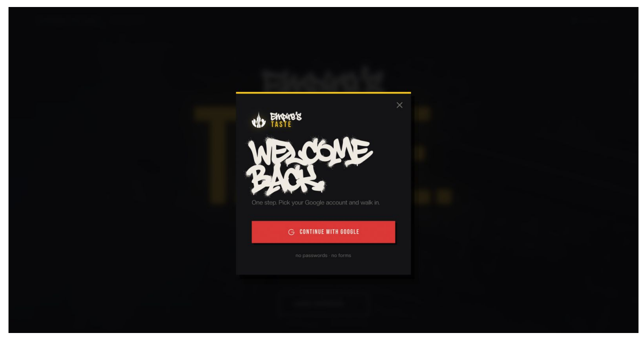

Fig. 6. Supplementary evidence of the Google authentication flow.

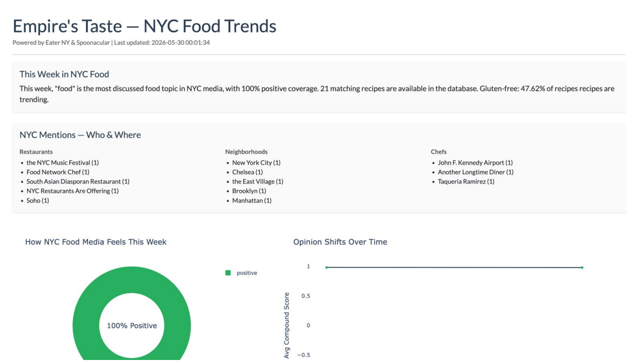

Fig. 7. Supplementary evidence of the sentiment analysis dashboard.

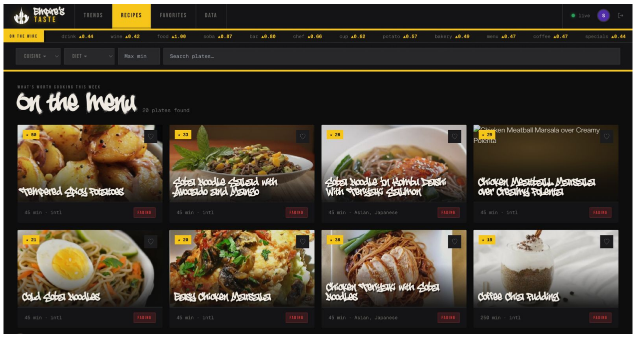

Fig. 8. Supplementary evidence of recipe recommendations generated from the analysis.

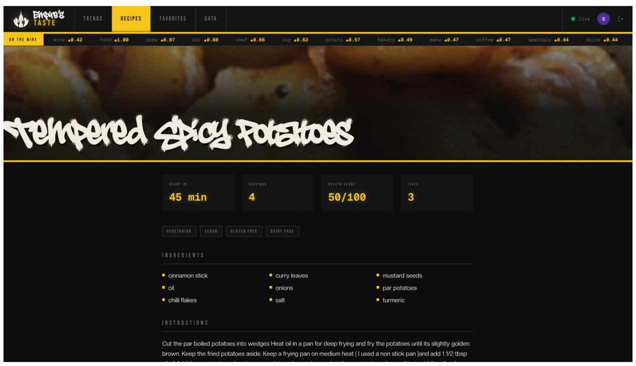

Fig. 9. Supplementary evidence of the detailed recipe view.

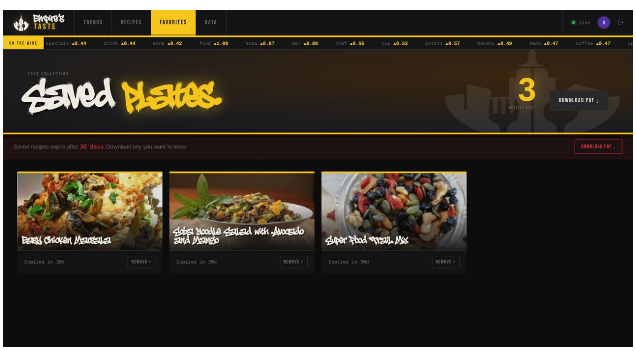

Fig. 10. Supplementary evidence of the favorite dishes section.

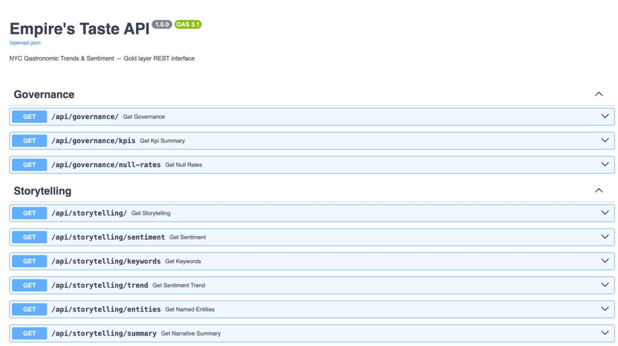

Fig. 11. Supplementary evidence of the API used for web application intercommunication.

The reproducibility check also confirms that the presentation layer is separated from the ingestion and cleaning layers. Both Dash applications use the same latest-partition loading strategy: find the newest Gold date folder, locate the expected Parquet pattern, and read it with Pandas. This prevents the dashboards from depending on raw JSON files, Silver intermediate schemas, or external API availability during visualization.

VII. Challenges and Resolutions

The main challenges in this workshop were related to integration reliability, Gold-layer compatibility, and visualization clarity. Table VIII provides a concise summary, followed by a more detailed explanation of the most important resolutions.

The first integration challenge was caused by container state. Recreating containers without rebuilding the image left the Airflow environment without the required PySpark dependency. This was resolved by using `docker-compose up --build -d` whenever Dockerfiles or dependency files changed. A related configuration issue occurred when updated API credentials were not reflected after a simple container restart. The solution was to recreate the services so the current `.env` file was read again at container startup.

The Gold transformation also required corrections before the dashboards could load the data reliably. The keyword sentiment aggregation failed because Spark Row-like objects were treated with dictionary-style access instead of attribute access. Correcting this access pattern allowed the Gold storytelling summary to be produced successfully. The keyword extraction output was also refined because generic terms that were not related to food appeared in the trend results. A food-domain filtering step was added so the storytelling dashboard would focus on culinary topics rather than unrelated words.

The visualization challenges were different from the pipeline challenges. The dashboards needed to be informative without becoming crowded, especially at a standard 1080p browser resolution. This was resolved by arranging content from summary to detail, using cards for headline values, using bar charts for ranked or multi-category comparisons, and reserving line charts for time-based trends. Color was standardized across both dashboards so that sentiment and quality signals remain consistent, while labels, legends, titles, and hover text provide meaning even when color alone is insufficient.

TABLE VIII
CONCISE CHALLENGES AND RESOLUTIONS.

Challenge Area	**Resolution**
Container dependencies	Rebuilt images to include PySpark and the required NLP model
Runtime configuration	Recreated containers so updated `.env` values were loaded
Gold compatibility	Corrected Spark Row access and validated dashboard fields
Visualization clarity	Used readable chart types, consistent colors, and responsive layout

VIII. RESULTS & DISCUSSION

The main result of this workshop is a working presentation layer that consumes only Gold Parquet files generated by the pipeline. The integration verification confirmed that the Airflow workflow can execute the required sequence from raw article ingestion to dashboard-ready Gold summaries. The governance and storytelling outputs were validated as separate files so that each dashboard can start independently and read only the dataset relevant to its purpose. This separation is important because a failure in one presentation view should not prevent the other dashboard from loading its own Gold data.

The governance dashboard provides evidence that pipeline quality can be assessed from a browser without manually opening Parquet files. Its KPI cards summarize record volume, completeness, duplication, and schema compliance, while the null, outlier, and volume charts provide more detailed diagnostic views. The data quality table complements the charts by preserving the complete governance record, allowing engineers to move from a visual warning to the exact field, source, and metric that produced it.

The storytelling dashboard demonstrates that the same Gold layer can also serve a non-technical audience. Instead of exposing raw NLP or scoring terminology, it presents a narrative summary, sentiment distribution, sentiment trend, top food keywords, source comparison, activity volume, and NYC mentions in plain language. This design supports the functional user's main questions: what food topics are being discussed, whether the tone is positive or negative, which names and places appear frequently, and whether recipe availability aligns with media attention.

The visualization design standards were also met across both dashboards. Charts use descriptive titles, labeled axes, legends when multiple series are present, and source annotations tied to the Gold layer. Time-based patterns are displayed with line charts, multi-category comparisons use bar charts, and the sentiment palette remains consistent across views. As a result, the dashboards are not only functional but also coherent, readable at standard screen resolutions, and suitable for inclusion as reproducibility evidence in the final report.

IX. CONCLUSIONS

Workshop 4 successfully delivered the dashboard layer of the Empire's Taste project. The pipeline was verified end to end, the Gold Parquet outputs were checked for dashboard completeness, and two independent Plotly Dash applications were documented: one for governance monitoring and one for narrative food-trend analysis. The implementation solves the workshop problem by connecting validated Gold data to browser-accessible interfaces that support both technical and functional users.

The work was successful because the dashboards are not isolated prototypes; they are integrated with the same reproducible data flow used by the rest of the project. The governance dashboard helps detect data quality issues, while the storytelling dashboard converts sentiment, keyword, entity, source, and activity summaries into usable insights for food professionals. Future work should focus on adding the final evidence captures, expanding the historical data window for richer trend analysis, and connecting the Gold serving layer to the planned API or mobile interface.

X. REPOSITORY

The full project source code, data samples, and documentation are available in the `Food-Gastronomy-Trends-Sentiment` GitHub repository.

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